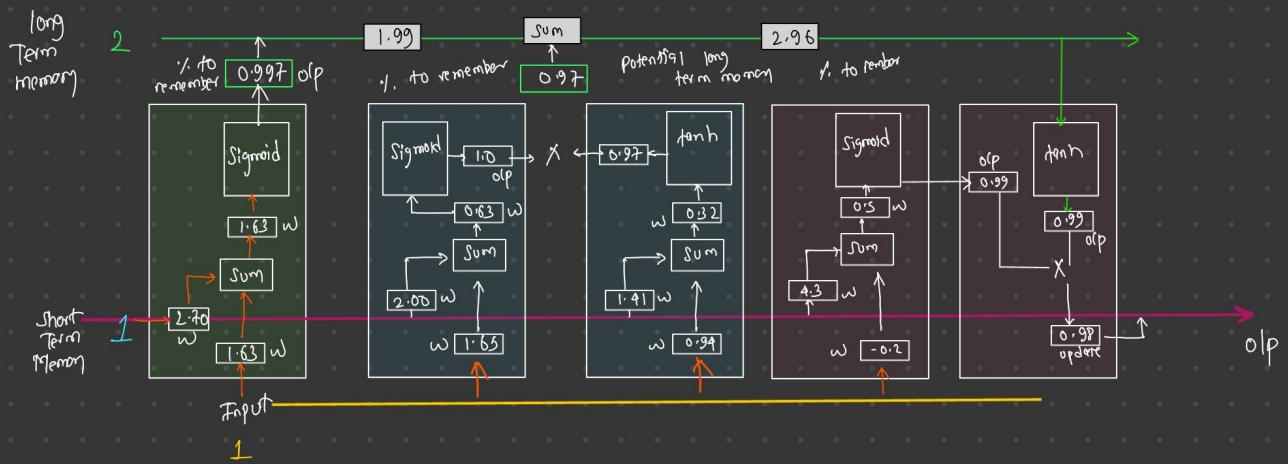




problem statement

Translation:

English

Spanish

Hi How are you

Hola como estas

Hi → Hola

How → como

are you → estás → eres

How are you → estás → feeling & condition

→ word 2vec

English → 1 word → 2

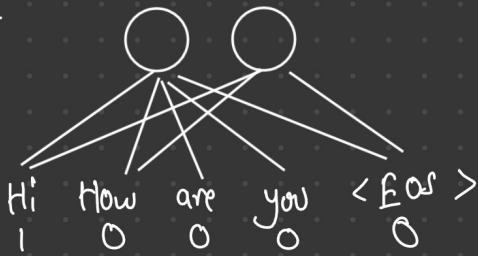
→ fasttext ←

Spanish → 1 word → 2

→ glove embedding ←

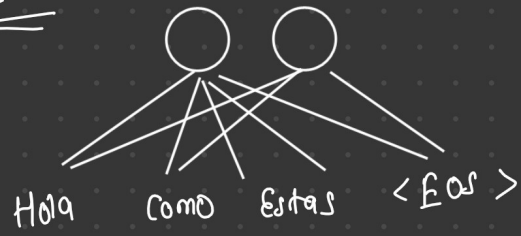
Hi → [0.437 0.238]

$\epsilon-1$



English - embedding

$\epsilon-2$



Spanish Embedding

Input $0.3, 0.5, -1.5, 2$

$\frac{0.3 \cdot 0.5}{e} + \frac{0.5 \cdot -1.5}{e} + \frac{-1.5 \cdot 2}{e} + \frac{2 \cdot 0.3}{e}$

Iteration-1 $\rightarrow$ Hola Como Estas <Eos>

Iteration-1 $\rightarrow$ Hola Como Estas <Eos>

Iteration-1 $\rightarrow$ Hola Como Estas <Eos>

Iteration-1 $\rightarrow$ Hola Como Estas <Eos>

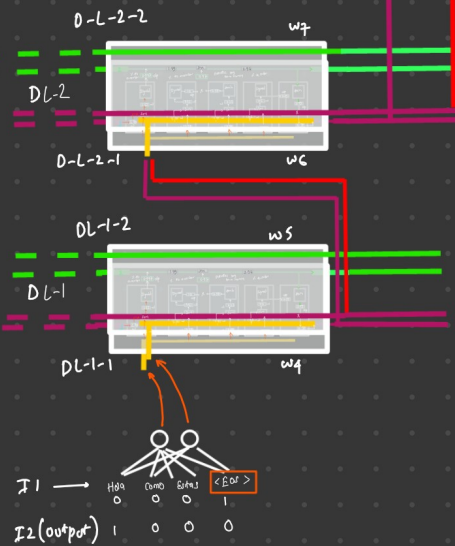
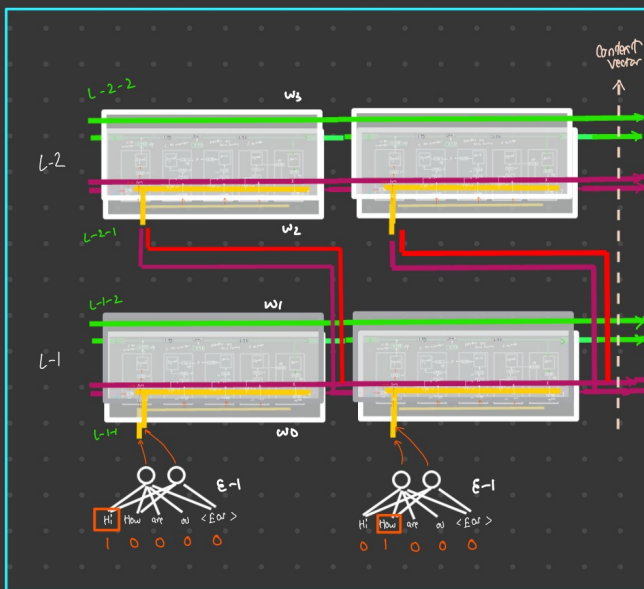
0.5 0.5 0.1 0.1

Softmax

0.3 0.5 -1.5 2

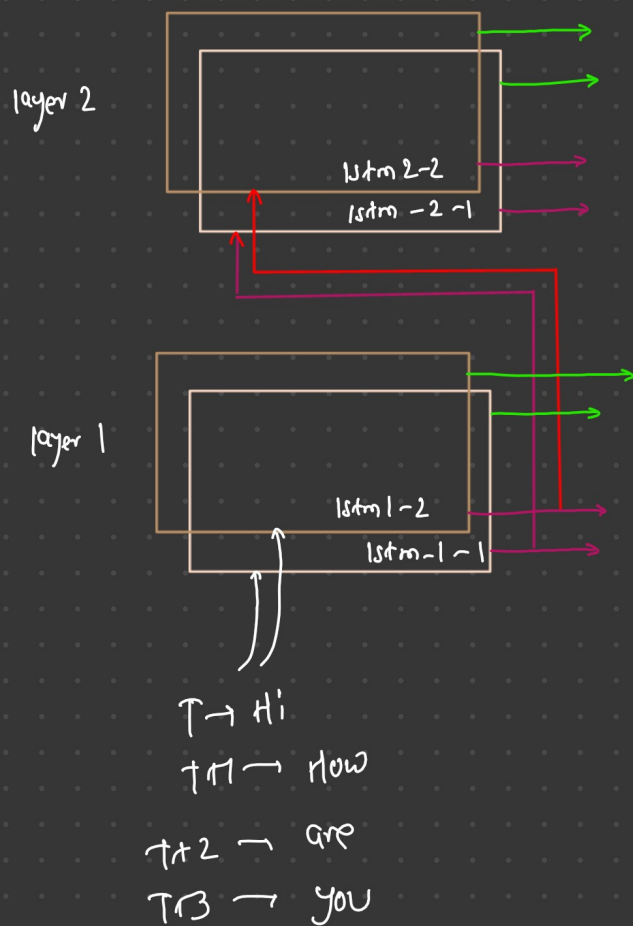
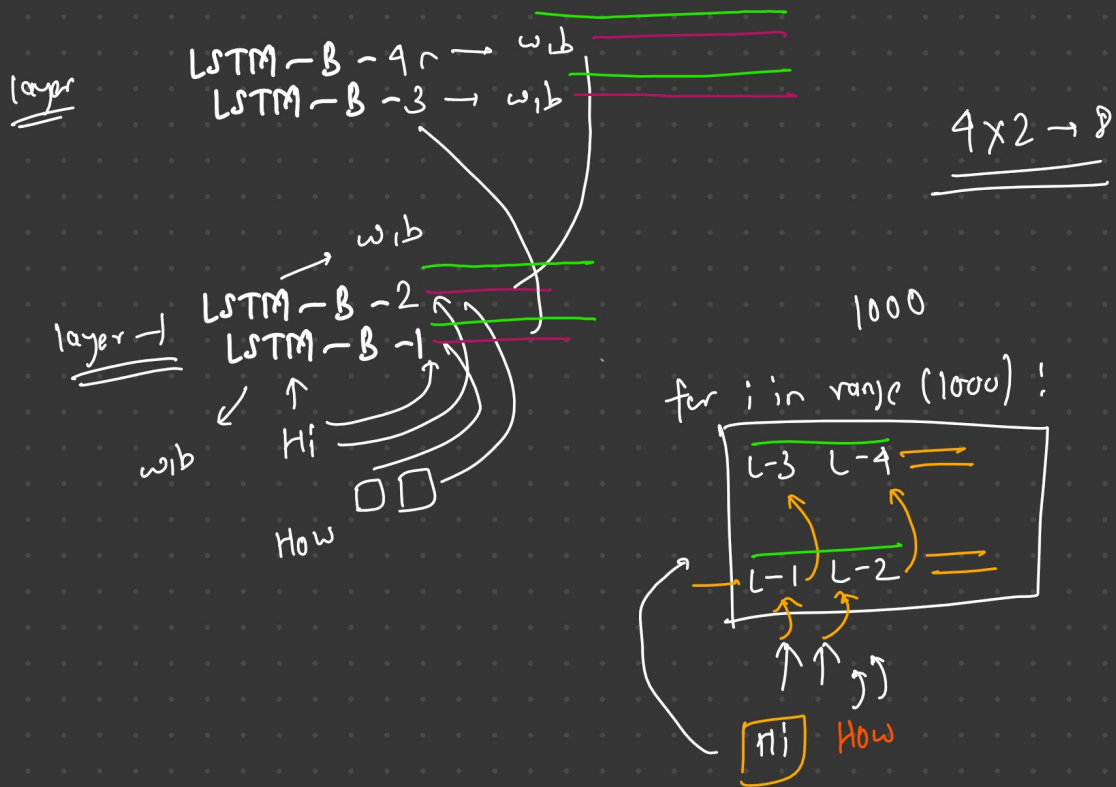
-0.5 1.2

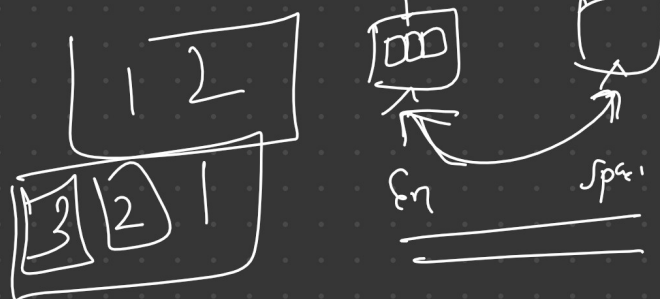
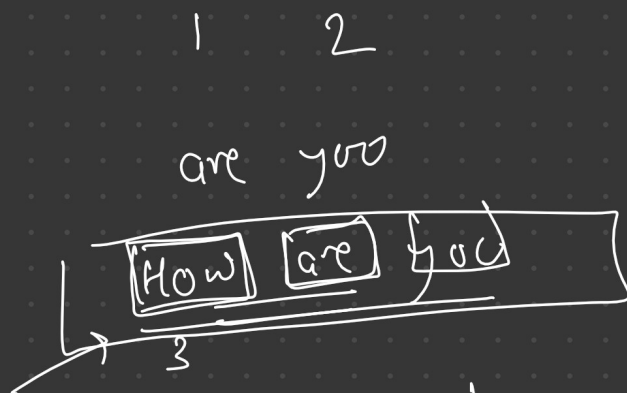
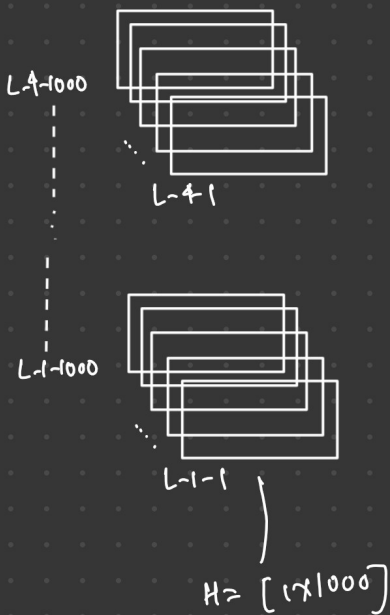
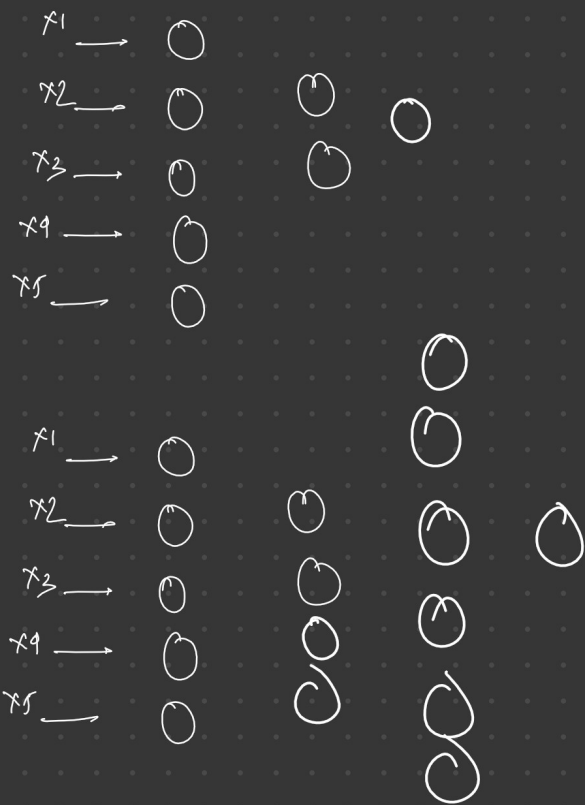
Encoder

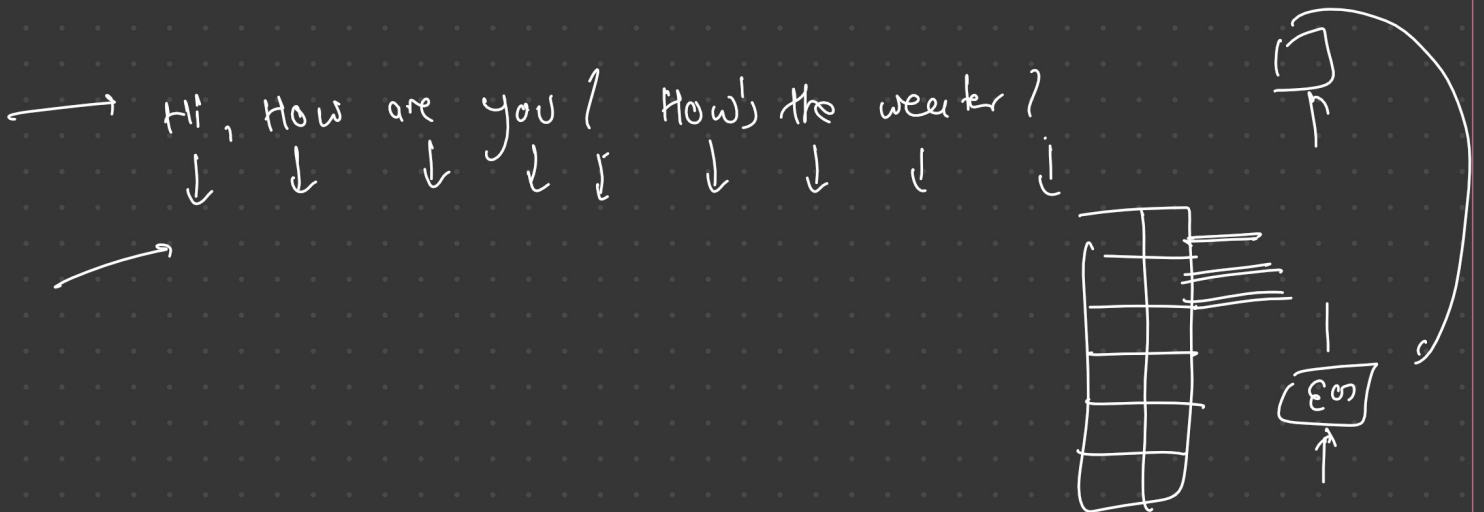
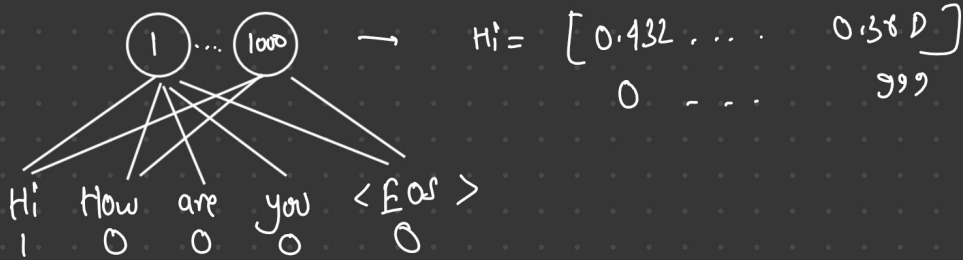


2 layer

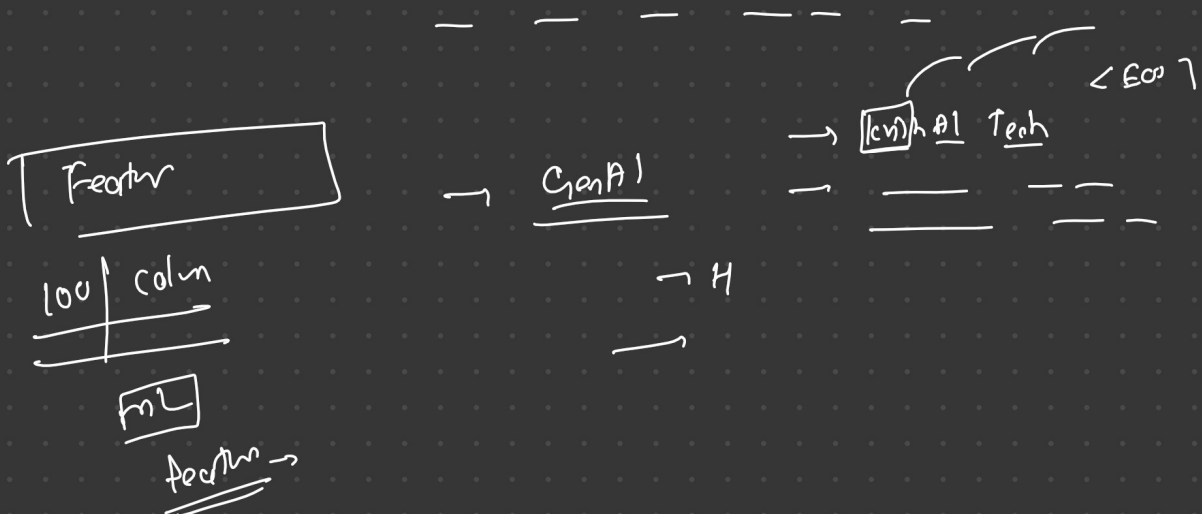
1 - 2 LSTM

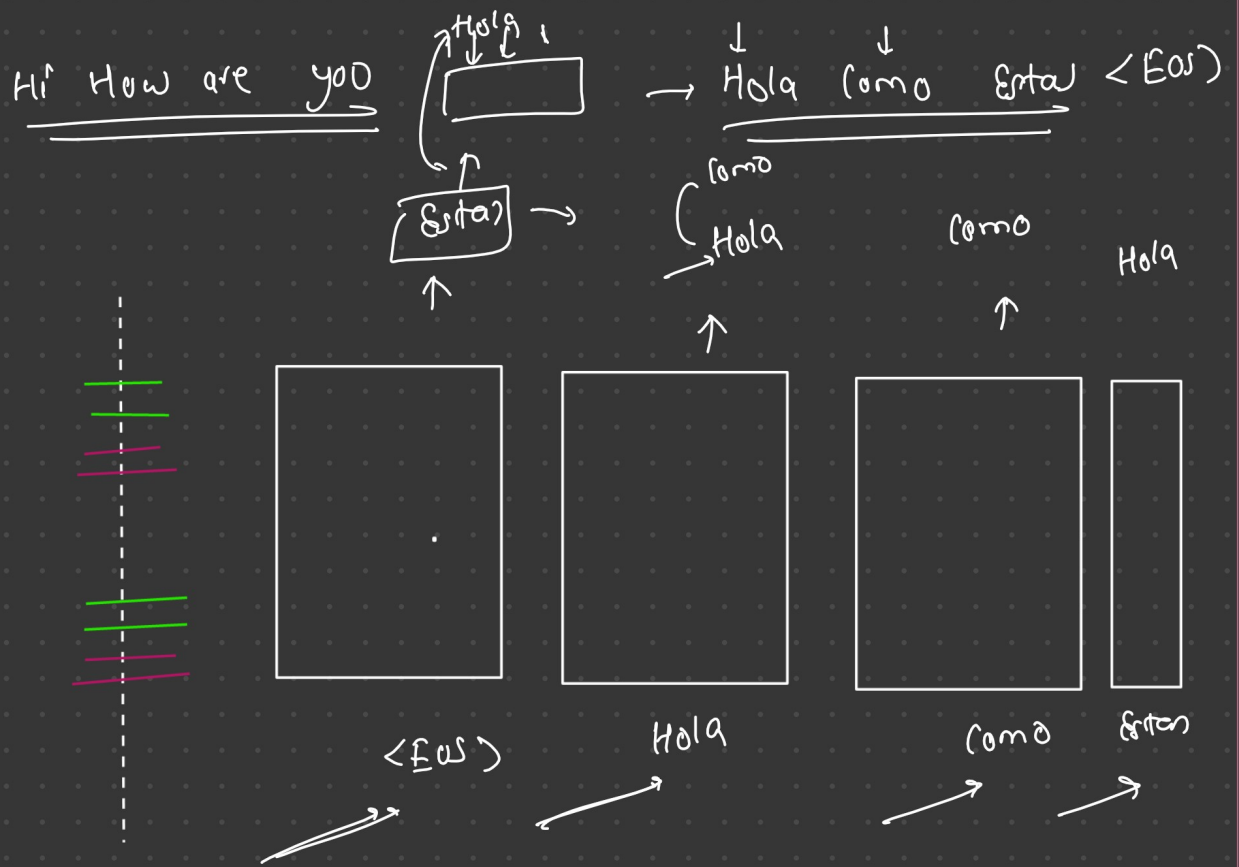






chatgpt → Query → Input → Encoder





Teacher forcing

AI → RL, ML, DL, GAN → find complex relation between numbers

